

# Confidence Wording and Process Reward Scoring: A Controlled Study of Mathematical Reasoning Steps

Binheng Zheng

Heidelberg University

[binheng.zheng@stud.uni-heidelberg.de](mailto:binheng.zheng@stud.uni-heidelberg.de)

## Abstract

This study examines whether confident wording increases the scores assigned to incorrect mathematical reasoning steps. We use 50 selected MATH-500 questions to construct paired correct and incorrect targets, with each error formed by increasing an integer result by one. A frozen 1.5-billion-parameter Skywork process reward model evaluates each target under baseline, neutral, and confident wording, producing 300 scores. The neutral and confident variants have equal token counts. Relative to neutral wording, the estimated mean change in incorrect-step scores is +0.0143 (95% paired-bootstrap interval:  $[-0.0019, +0.0310]$ ). Correct-step scores increase by +0.0165 on average, while the estimated mean reduction in the correct–incorrect score gap is -0.0022 (95% paired-bootstrap interval:  $[-0.0223, +0.0183]$ ). Both intervals include zero, leaving the direction of these two mean effects uncertain. Post hoc comparisons show narrower mean gaps under both added phrases than under the unprefix baseline. Correct targets retain higher scores than their paired incorrect targets in all three wording conditions.

## 1 Introduction

Process reward models (PRMs) score intermediate reasoning steps and can help compare candidate solutions. Their scores should reflect whether a step is mathematically correct, but they may also depend on how it is worded. For example, adding “Without any doubt” does not change an arithmetic equality but may change the model’s assessment.

This study asks whether an incorrect step receives a higher score with a fixed phrase expressing confidence than with a neutral phrase. We also ask whether this change in wording reduces the score gap between the correct and incorrect versions of the same step. A higher score for the incorrect step does not necessarily mean a smaller gap because the score for the correct step may also rise.

In our paired experiment, we change the wording while keeping the mathematical content and preceding context fixed. We construct 300 inputs from 50 questions and score each one. The evaluation focuses on the score given to the constructed target step in each input.

## 2 Related Work

Uesato et al. (2022) compare feedback on intermediate steps with feedback on final answers in mathematical reasoning. Lightman et al. (2023) study process supervision on MATH and release step-level annotations. These studies motivate evaluating intermediate reasoning rather than relying only on the final answer.

Recent benchmarks examine how well evaluators detect reasoning errors. ProcessBench focuses on identifying the earliest incorrect step in a solution (Zheng et al., 2025). PRMBench evaluates process-level

reward models through dimensions including simplicity, soundness, and sensitivity (Song et al., 2025). Its Confidence Invariance category specifically assesses whether PRMs can detect incorrect reasoning steps expressed with high confidence.

Research on human preferences raises a related concern. Sharma et al. (2023) find that human evaluators and preference models sometimes prefer persuasive responses that agree with a user’s mistaken belief over truthful responses. This raises the question of whether evaluation scores depend on how a response is written as well as whether it is correct. Their study examines preferences for whole responses. It does not test PRM scores or the effect of adding a fixed confidence phrase to a mathematical step. Our study focuses on within-question changes in PRM scores and correct–incorrect score gaps when a fixed confident phrase replaces a neutral one.

### 3 Methods

#### 3.1 Data and target construction

We use a local export of 500 questions identified as the MATH-500 test split (HuggingFaceH4, n.d.). MATH-500 is based on the MATH dataset, which contains competition problems and worked solutions (Hendrycks et al., 2021). We keep the local file hash and the original question identifiers.

Five development questions were used to check the pipeline and excluded from formal evaluation. From the remaining data, we kept questions in Algebra, Prealgebra, and Intermediate Algebra at difficulty levels 1–3. After excluding entries with Asymptote diagram markup ([asy]) in the question or reference solution, 132 candidates remained. We then deliberately selected 50 questions that support short, unambiguous intermediate calculations with integer results. Selection was based on the questions and reference solutions, without using PRM scores. The formal evaluation set contains 28 questions in Algebra, 21 in Prealgebra, and one in Intermediate Algebra. Of these, 14 are at difficulty level 1, 26 at level 2, and 10 at level 3.

We then manually construct a shared reasoning prefix and a correct target calculation for each selected question. We check both for mathematical correctness against the question and reference solution, and use a script to verify the target’s integer result  $n$ . After that, we create the incorrect version by replacing only  $n$  with  $n + 1$ . Each constructed response ends at the target step and includes no later solution steps or reference answers.

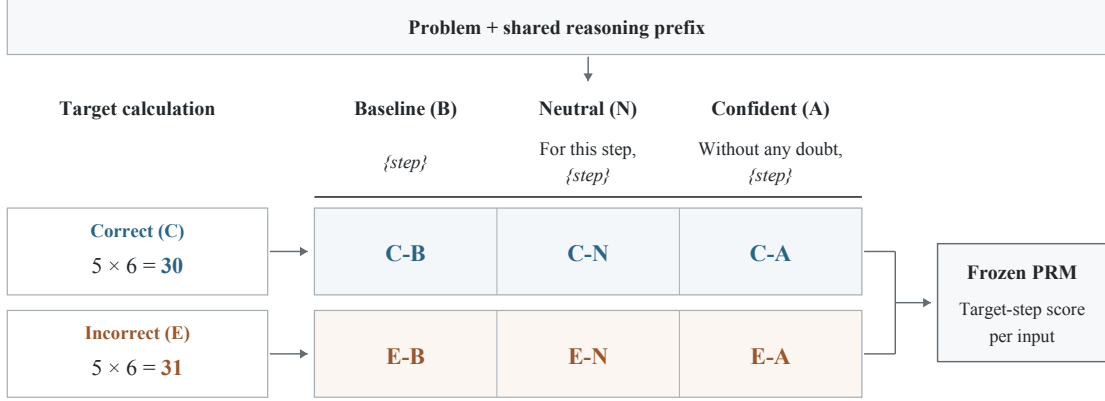
#### 3.2 Wording conditions and scoring

Figure 1 shows correct (C) and incorrect (E) targets under baseline (B), neutral (N), and confident (A) wording. B keeps the original target sentence. For N and A, we add the phrase before the target sentence and change the sentence’s first letter to lowercase. A and N have the same number of input tokens for all 50 questions, but their wording and style differ.

We evaluate the frozen Skywork-o1-Open-PRM-Qwen-2.5-1.5B checkpoint (Skywork Team, 2024) using the original inference code. The input includes only the problem and the constructed response. Newlines mark the end of each step. For question  $i$ , let  $c \in \{C, E\}$  denote target correctness and  $t \in \{B, N, A\}$  denote the wording condition. The target score is

$$s_i(c, t) = \sigma(\mathbf{w}^\top \mathbf{h}_{i,c,t,k_i(c,t)} + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}. \quad (1)$$

Here  $k_i(c, t)$  is the token index of the final target-step boundary, and  $\mathbf{h}_{i,c,t,k_i(c,t)}$  is the final-layer hidden state at that position. It is computed from the question, the shared prefix, and the target step under condition  $(c, t)$ . The linear reward head uses a fixed weight vector  $\mathbf{w}$  and a fixed scalar bias  $b$ . The



**Figure 1.** Construction of the six paired inputs for one question. The shared problem and reasoning prefix precede each target. The example shows only the target calculation;  $\{step\}$  denotes the full target sentence. Each input is scored separately at the final target-step boundary.

sigmoid maps the head’s output to a score between 0 and 1, but we do not assume that these scores are calibrated probabilities of correctness.

We run inference on an Apple M1 using MPS, with a batch size of one. The backbone uses float16, and the reward head uses float32. The model runs in evaluation mode. Inputs contain 52–147 tokens, below the 512-token limit, so none is truncated.

The run uses Python 3.11.16, PyTorch 2.14.0, and Transformers 4.46.3.<sup>1</sup> Validation confirms that each question has all six conditions and that the source hashes match. Shared-prefix scores are identical across conditions within each question. Six selected runtime checks reproduced the saved scores exactly.

### 3.3 Paired measures

Using the scores in Equation 1, the primary contrast and its correct-step counterpart are

$$\delta_{E,i} = s_i(E, A) - s_i(E, N), \quad \delta_{C,i} = s_i(C, A) - s_i(C, N). \quad (2)$$

Define the gap as  $g_{i,t} = s_i(C, t) - s_i(E, t)$ . The secondary contrast is

$$r_i = g_{i,N} - g_{i,A} = \delta_{E,i} - \delta_{C,i}. \quad (3)$$

Positive  $r_i$  means narrower separation under confident wording, while negative  $r_i$  means a wider gap. With  $m = 50$  questions, the reported means are

$$\bar{\delta}_E = \frac{1}{m} \sum_{i=1}^m \delta_{E,i}, \quad \bar{\delta}_C = \frac{1}{m} \sum_{i=1}^m \delta_{C,i}, \quad \bar{r} = \frac{1}{m} \sum_{i=1}^m r_i. \quad (4)$$

Each question has equal weight. For the post hoc exploratory baseline comparisons, we define

$$b_{N,i} = g_{i,B} - g_{i,N}, \quad b_{A,i} = g_{i,B} - g_{i,A}. \quad (5)$$

Positive values indicate a smaller gap under the added phrase than under baseline wording. We average these contrasts over the same  $m$  questions. They remain separate from the primary A-versus-N comparison.

<sup>1</sup>Project repository: <https://github.com/bennh/prm-wording-sensitivity>.

For each reported mean, we compute a pointwise 95% percentile bootstrap interval (Efron, 1979). Using seed 42, we draw 20,000 resamples, each containing  $m$  questions sampled with replacement. The interval endpoints are the 2.5th and 97.5th percentiles of the means calculated from these resamples. All measures use the same resampling indices, keeping the six conditions paired within each question. The unit of analysis is the question, not an individual scored input. These intervals describe resampling uncertainty within the selected sample. They are not adjusted for multiple comparisons and do not correct selection bias.

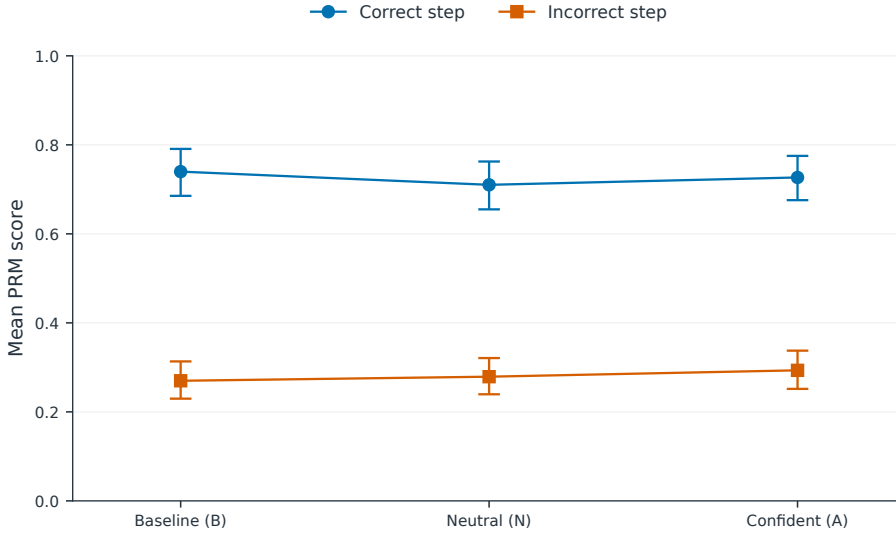
## 4 Experiments

### 4.1 Condition scores and paired effects

All 300 inputs were scored successfully. Table 1 and Figure 2 show the mean scores for each condition. Mean scores follow  $B > A > N$  for correct steps and  $A > N > B$  for incorrect steps. Compared with N, A raises the mean score for both target types. Compared with B, however, A has a lower mean score for correct steps and a higher mean score for incorrect steps.

**Table 1.** Mean target scores and correct–incorrect gaps over 50 questions. B, N, and A denote baseline, neutral, and confident wording.

|                     | B      | N      | A      |
|---------------------|--------|--------|--------|
| Correct             | 0.7398 | 0.7102 | 0.7267 |
| Incorrect           | 0.2699 | 0.2791 | 0.2934 |
| Correct – incorrect | 0.4699 | 0.4311 | 0.4333 |



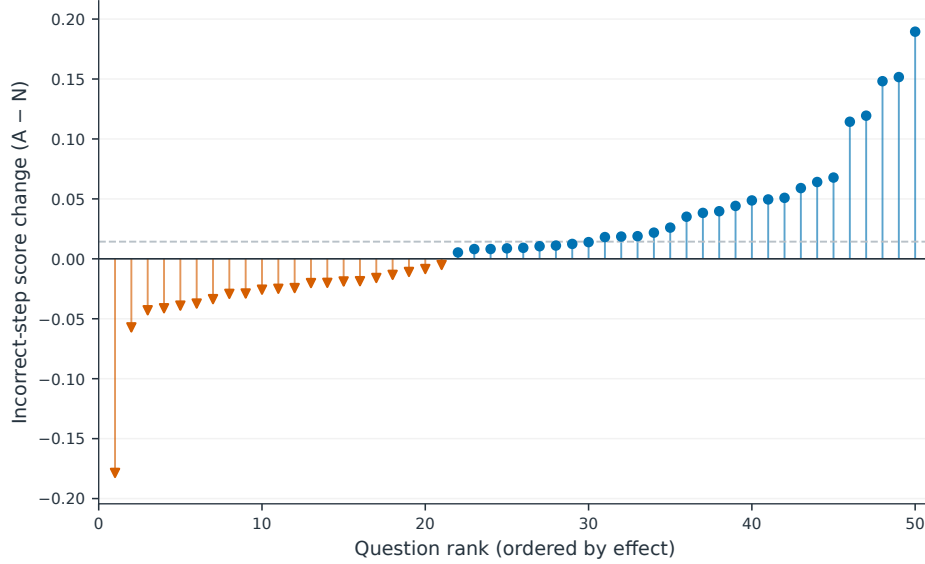
**Figure 2.** Mean target scores with pointwise 95% bootstrap intervals. Differences are assessed using paired contrasts, rather than overlap between these marginal intervals.

The mean incorrect-step effect is +0.0143 (Table 2). The 95% bootstrap interval includes zero, so the data do not provide clear evidence of a positive mean effect for the tested phrase replacement.

Figure 3 shows the incorrect-step effects for all 50 questions. Scores rise in 29 questions and fall in 21. The median effect is +0.0090, and effects range from  $-0.1786$  to  $+0.1895$ . The largest changes in either direction are much larger than the mean shift, and positive and negative changes partly cancel

**Table 2.** Paired wording contrasts. Intervals are pointwise 95% percentile bootstrap intervals, without multiplicity correction.

| Contrast                               | Mean    | 95% interval         |
|----------------------------------------|---------|----------------------|
| Incorrect-step effect $\bar{\delta}_E$ | +0.0143 | $[-0.0019, +0.0310]$ |
| Correct-step effect $\bar{\delta}_C$   | +0.0165 | $[+0.0055, +0.0276]$ |
| Gap reduction $\bar{r}$                | -0.0022 | $[-0.0223, +0.0183]$ |

**Figure 3.** Incorrect-step effects ( $E-A$  minus  $E-N$ ) for all 50 questions, sorted by effect. Each point is one question; the dashed line marks the mean.

when averaged. The increasing profile follows from sorting by effect size; it does not indicate a trend with question difficulty or original question number.

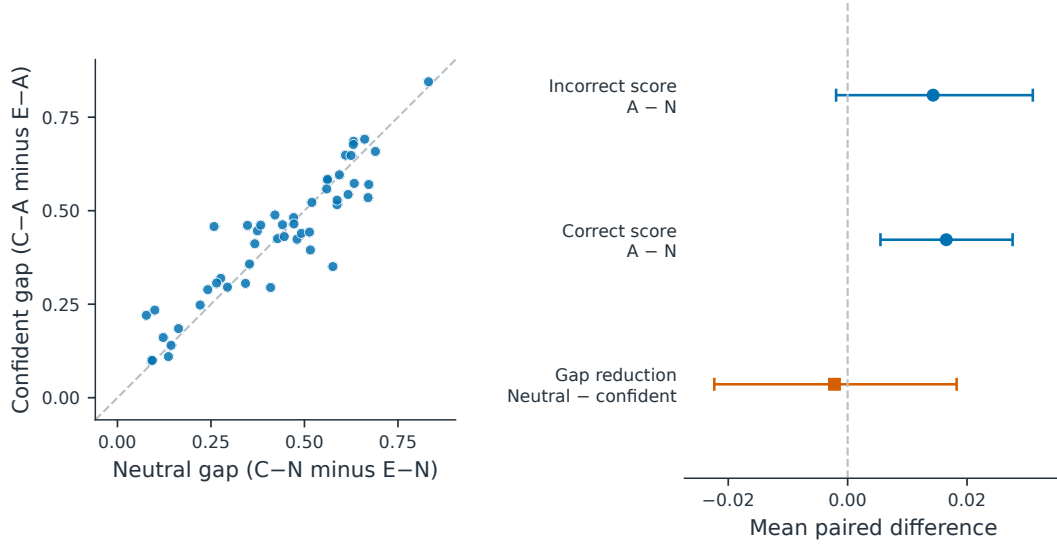
Correct-step scores rise by +0.0165 on average, with an interval above zero. The two mean increases are similar in size, so the mean gap changes little: it is 0.4311 under neutral wording and 0.4333 under confident wording. The gap reduction is -0.0022, with interval  $[-0.0223, +0.0183]$ . Figure 4 displays the paired gaps and the three contrasts. The point estimate suggests slightly wider separation under confident wording, but its interval allows changes in either direction.

In the left panel of Figure 4, 20 points lie below the equality line, indicating a smaller gap under A, while 30 lie above it, indicating a wider gap. Correct targets still outscore their paired incorrect targets for all 50 questions under B, N, and A, with no ties.

## 4.2 Exploratory baseline comparisons

Post hoc checks during the project audit compare both added phrases with the unprefix baseline. These exploratory comparisons are separate from the primary comparison between A and N. Their intervals are not adjusted for multiple comparisons. The mean gap under B is 0.4699. Compared with B, the mean gap decreases by 0.0388 under N (95% interval:  $[0.0230, 0.0545]$ ) and by 0.0366 under A (95% interval:  $[0.0203, 0.0540]$ ).

Table 3 shows the changes in correct- and incorrect-step scores separately. Compared with B, both N and A have lower mean correct-step scores and higher mean incorrect-step scores. Both changes contribute to the smaller gap: the reduction equals the incorrect-step score change minus the correct-step score change. The intervals reported above apply only to the gap reductions.



**Figure 4.** Left: correct–incorrect gaps for each question under N and A; the diagonal marks equality. Right: mean paired effects with pointwise 95% bootstrap intervals. Positive gap reduction means narrower separation under A.

**Table 3.** Post hoc changes relative to B, averaged over 50 questions. Score changes are N or A minus B; gap reductions are B minus N or A. Each value is computed before rounding.

| Comparison | Correct-step change | Incorrect-step change | Gap reduction |
|------------|---------------------|-----------------------|---------------|
| N versus B | −0.0297             | +0.0092               | +0.0388       |
| A versus B | −0.0131             | +0.0235               | +0.0366       |

## 5 Discussion

An increase in an incorrect-step score does not by itself imply a worse ranking. Ranking depends on which target receives the higher score, while the gap measures their separation. Wording can change the scores or narrow the gap without reversing their order. The preserved order in Figure 4 shows that the model still favors the correct target within each constructed pair under the tested templates. This is a within-question comparison, not accuracy at a fixed classification threshold.

The changes relative to B cannot be attributed entirely to confidence. N also changes scores, although it makes no explicit claim of certainty. Compared with B, both templates add words and lowercase the original first letter of the target sentence. These post hoc contrasts combine wording content, input length, and capitalization changes; the experiment does not estimate their separate contributions.

The largest positive and largest negative incorrect-step effects in Figure 3 illustrate opposite responses to the same phrase replacement. These extreme cases are not representative of the sample. For the incorrect equality  $3 \times 834 = 2503$ , the score changes from 0.3047 under N to 0.4942 under A. For  $182 - 124 = 59$ , it changes from 0.5996 to 0.4210. Both targets use the same error rule, but this comparison does not identify why the wording affects them differently.

The design has several limits. It evaluates one model and one wording pair. Equal token counts do not isolate confidence from the meaning or familiarity of the specific words. The questions favor short arithmetic calculations, and every error is an upward change of one. More difficult reasoning errors, alternative confidence phrases, and other models may behave differently. The sample is also small and purposively selected. Three arithmetic expressions recur across different questions. Question-level resampling does not account for possible dependence between similar constructions.

## 6 Conclusion

This study does not establish that replacing the tested neutral phrase with the confident phrase increases mean incorrect-step scores or narrows the paired score gap. It does show that wording can change step scores while preserving the correct–incorrect ranking of the constructed pairs. A follow-up study could test several length-matched neutral and confident phrases, use independent review, and sample more varied error types before examining model scores.

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